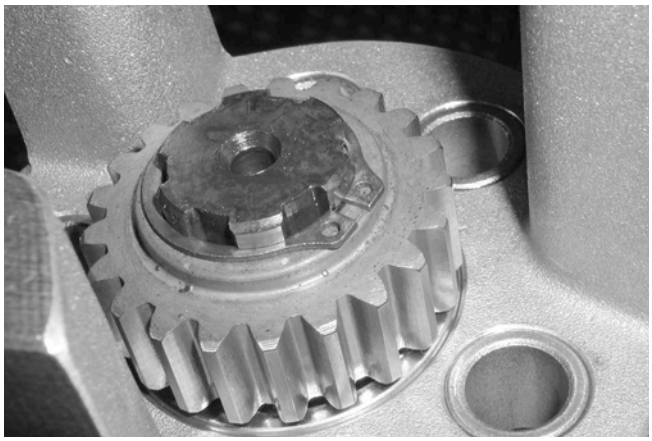
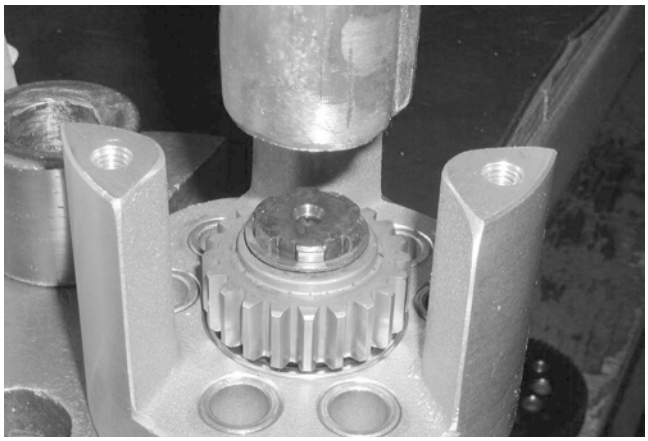
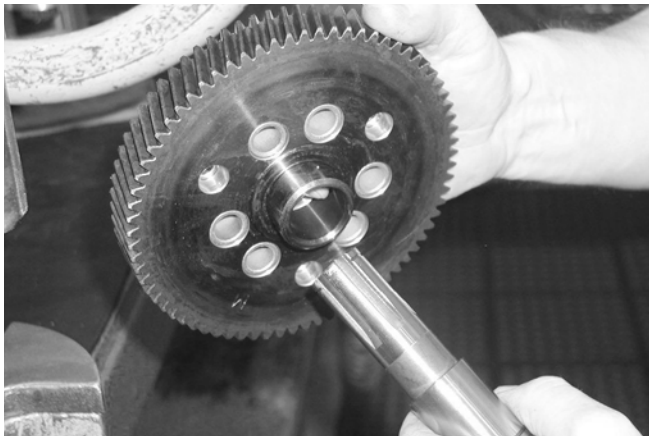
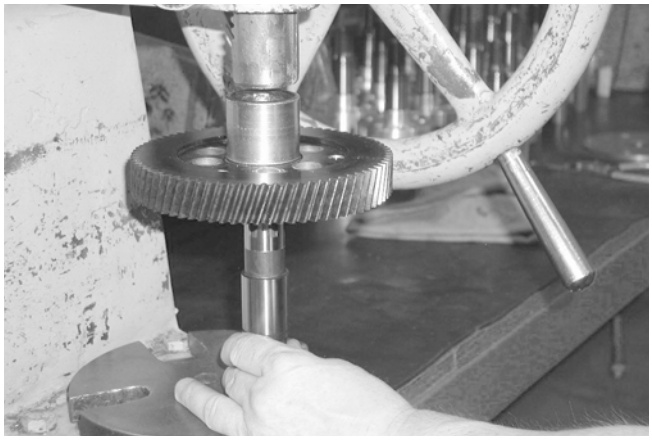
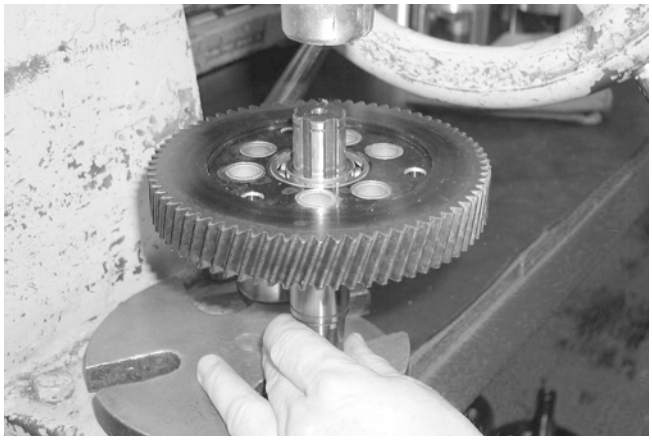


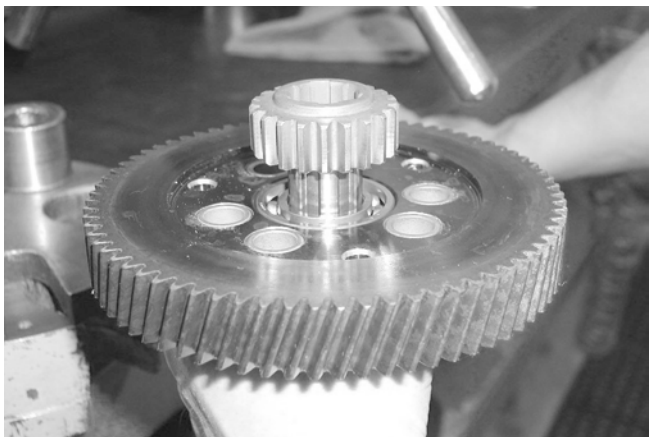


8. Wipe down the 2nd 34-98 Output Shaft and assemble to the Gear Plate using the same method as described for the Aluminum Cage.



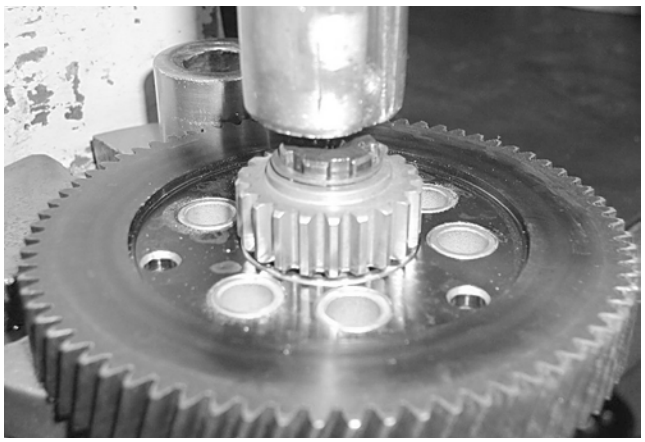


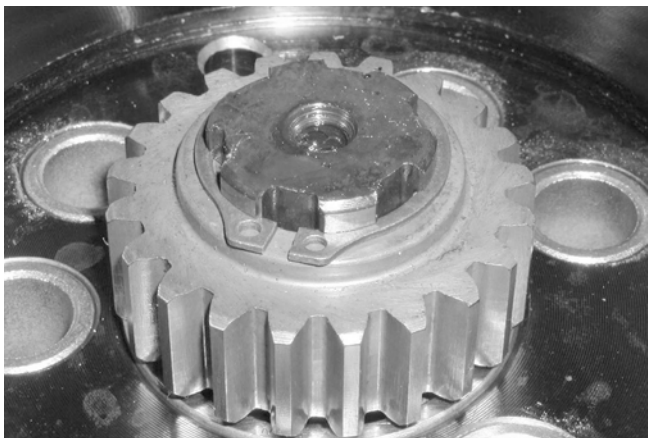


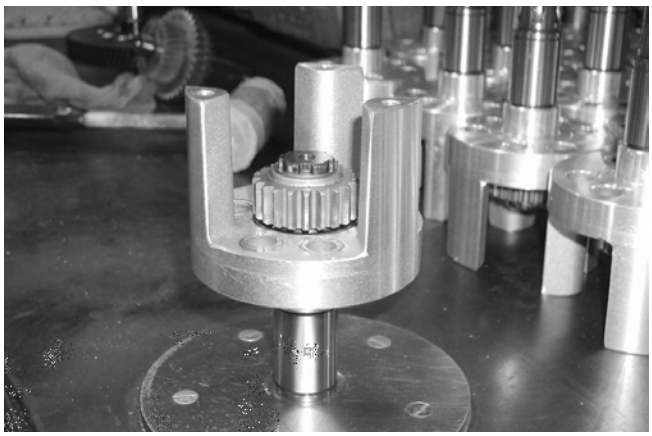


9. Orient the 2nd 34-14 Sun Gear as in instruction 8 and install and secure it with a 107-04 Retaining Ring using the same method as described for the Aluminum Cage in instruction 10.



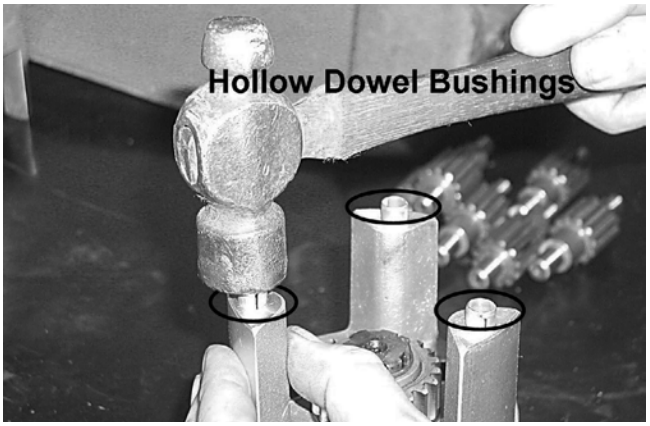


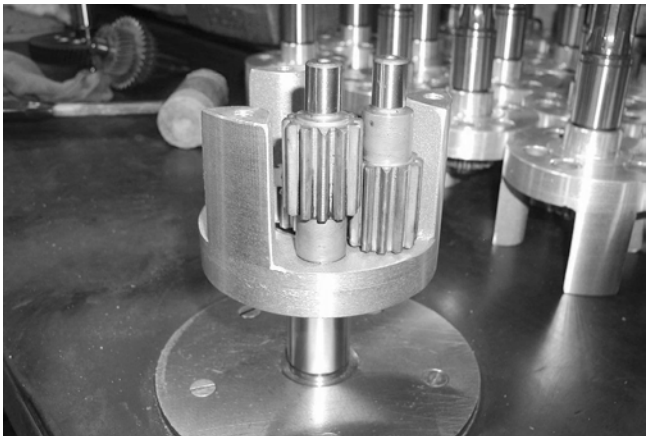




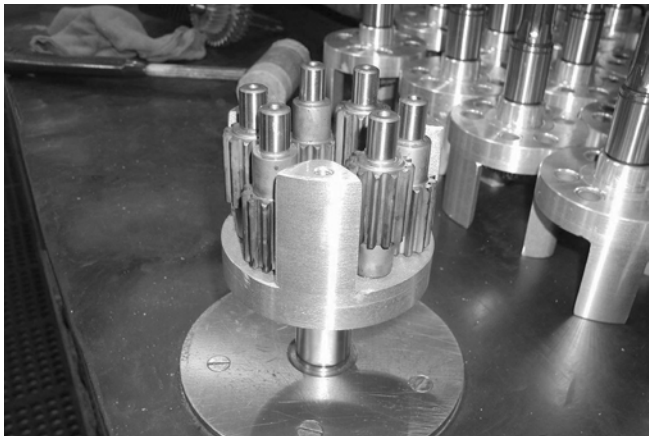
10. Set the assembled 34-122 Aluminum Cage into an assembly stand.

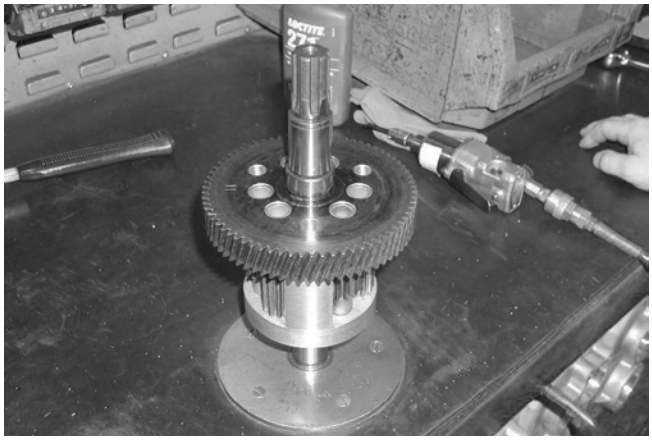
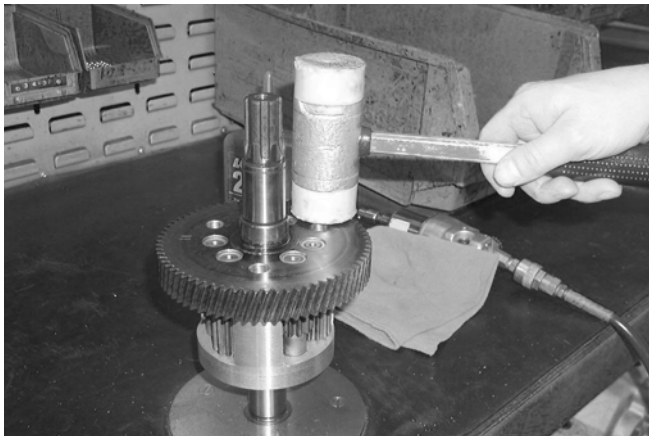
11. Install (3) 105-23 Hollow Dowl Pins into the mounting holes of the aluminum cage.





12. Install (6) 34-13 Pinions into the Oilite Bushings of the Aluminum Cage alternating the Pinions in an up and down manner.

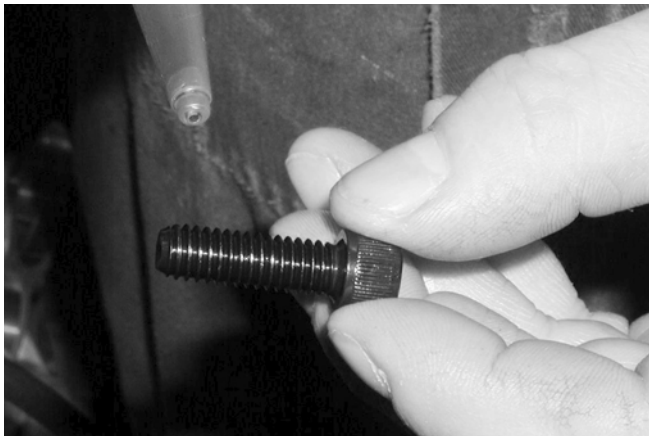


13. Assemble the Gear Plate to the Aluminum Cage. Align the top of the Pinions in the Cage, with the Oilite Bushings in the Gear Plate.

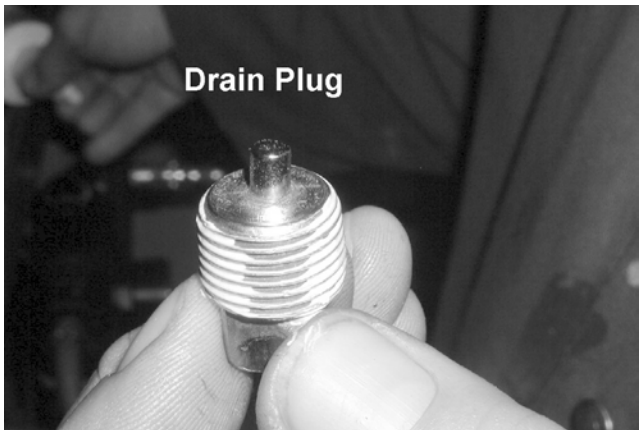
14. Seat the Gear Plate to the Aluminum Cage using a soft faced mallet.

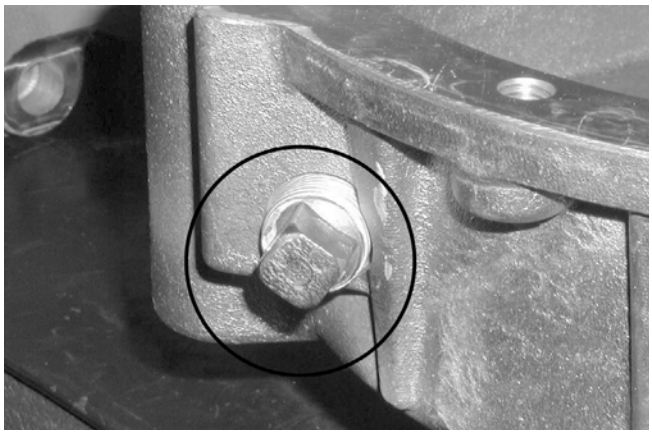
15 Apply Red 271 LOCTITE to each of the (3) 112-143 fasteners, and **torque down to 35 FT/LBS.**




Assembling Transmission Components to Housing

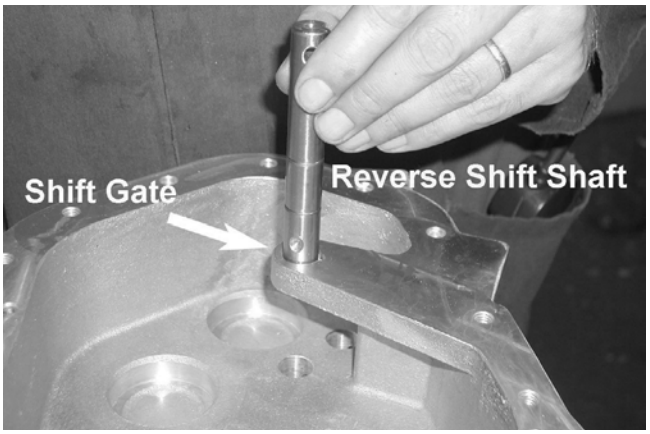
1. Apply some LOCTITE pipe sealant to the threads of the 103-02 drain plug and install into the housing.

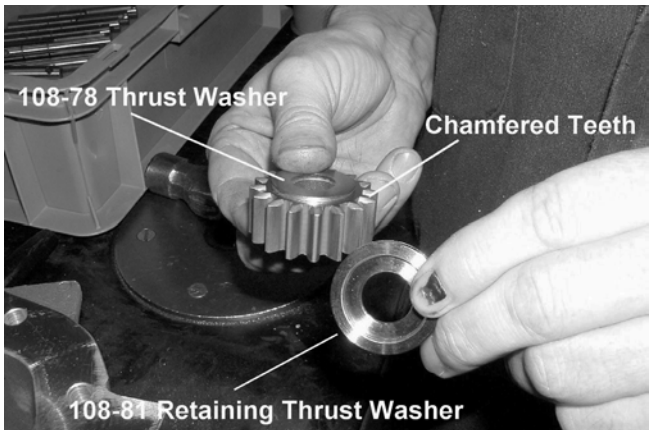




2. Start the 34-114 Reverse Shift Shaft into the housing boss.

3. Place a 108-78 Thrust washer above the 34-108 Reverse Idler, and a 108-81 Retaining Thrust Washer (machined side down), below the Reverse Idler. Push the 34-114 Reverse Shift Shaft completely through the Reverse Gear and washers and into the opposite side of the housing. Install the Reverse gear with the chamfered teeth facing up.

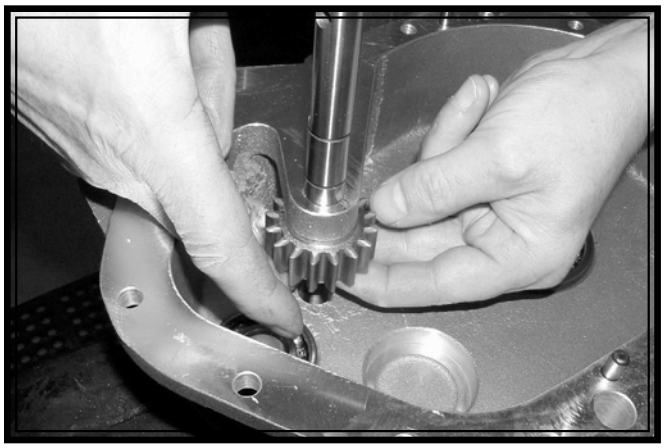


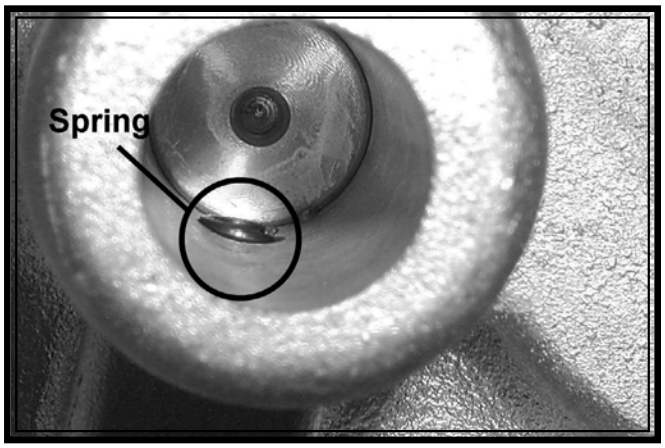


4. Install the 34-32 Spring into the machined hole of the 34-102 Right Hand Housing at the Reverse Shift Shaft location.

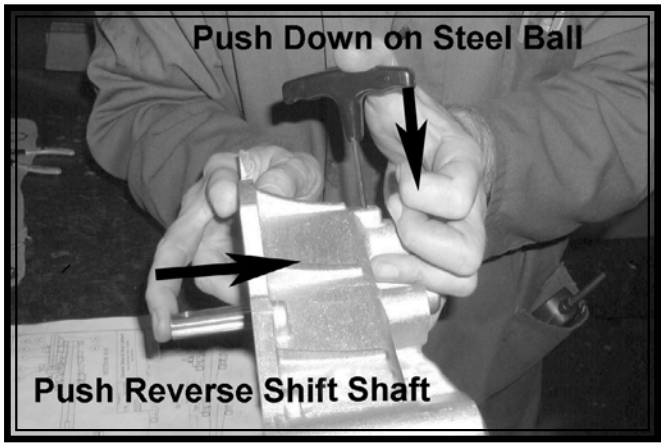
5. Install the 109-05 Steel Ball to the top of the Spring.

6. Push down on the Steel Ball using the access hole from the top of the Transmission Housing. While the ball is compressed, press the shaft over top of the ball.







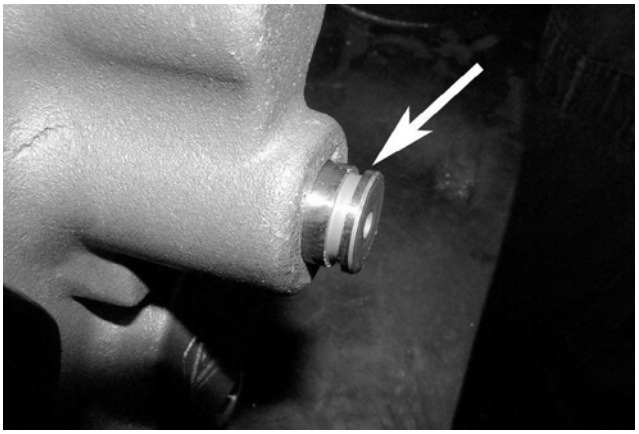


IMPORTANT

O-Ring installed to the machined groove located near the end of the reverse shift shaft and require these extra steps to install it.

i Continue to push the reverse shift shaft through the housing until the machined groove for the O-ring location is visible to the outside of the housing.

ii Install the 113-16 O-ring to the groove and apply some oil lubrication.

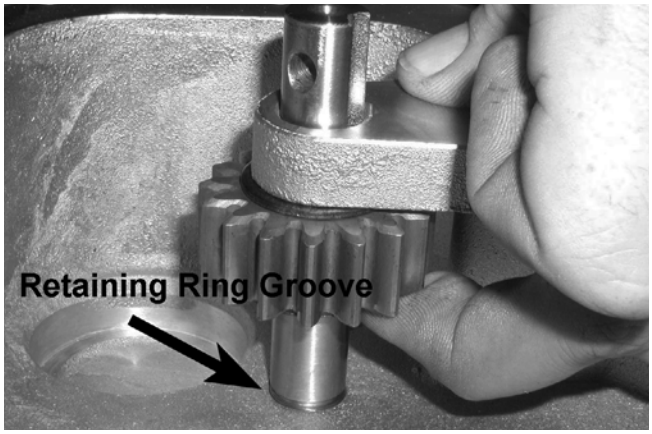


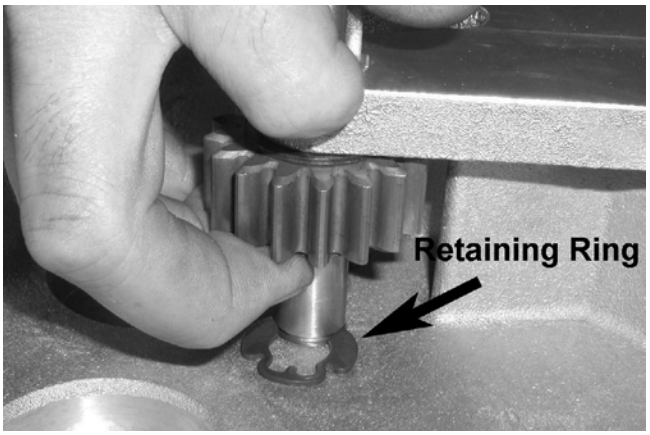


iii Push the reverse shift shaft back through the housing ensuring that you don't damage the O-ring as re-enters the housing.

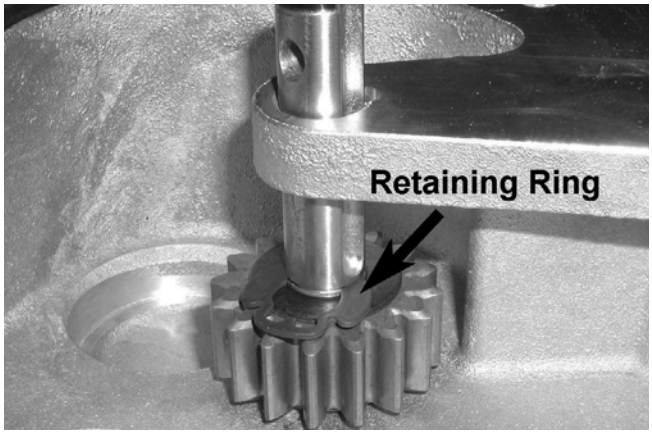
7. Pull the Reverse Shift Shaft up until the lower retaining ring groove location is visible. shaft is

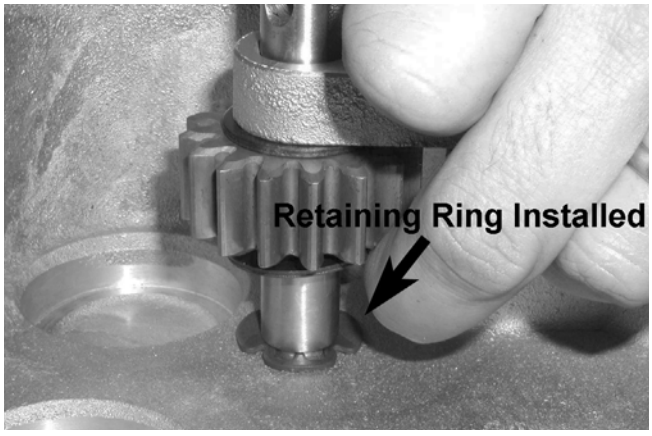
8. Install the 107-11 Retaining Ring onto the Reverse Shift Shaft.



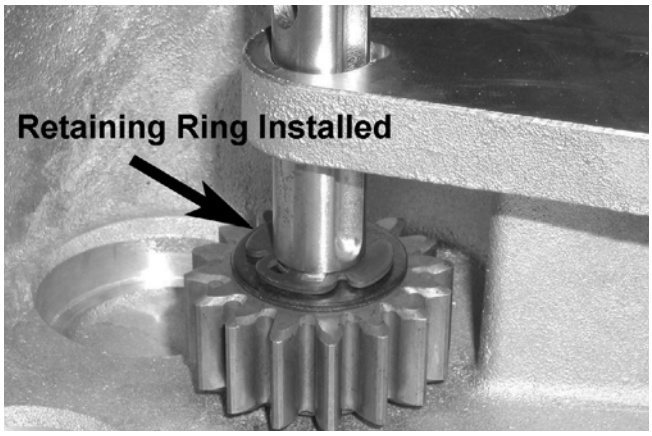


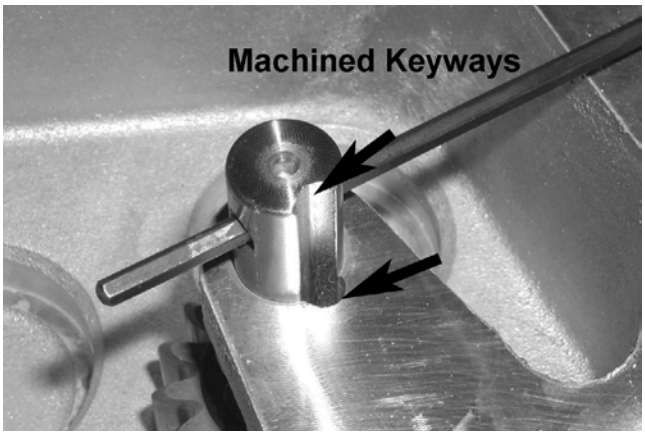
9. Lower the Reverse Idler Gear and install the second 107-11 Retaining Ring to the machined groove at the top of the Reverse Idler Gear.





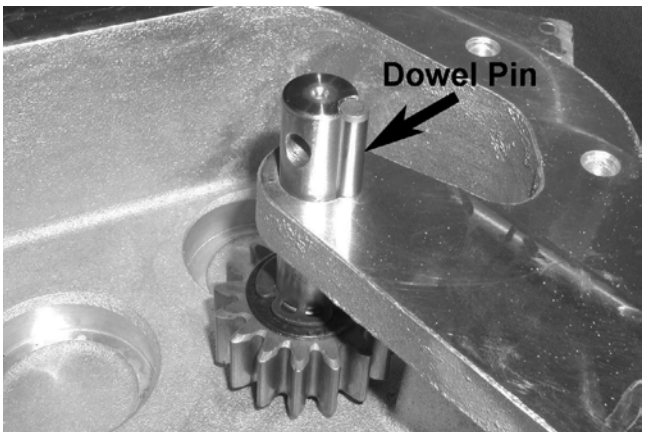
10. Insert a tool, eg. an allen wrench, into the machined hole of the Reverse Shift Shaft and turn the shaft until the keyways of both shaft and transmission boss are aligned.

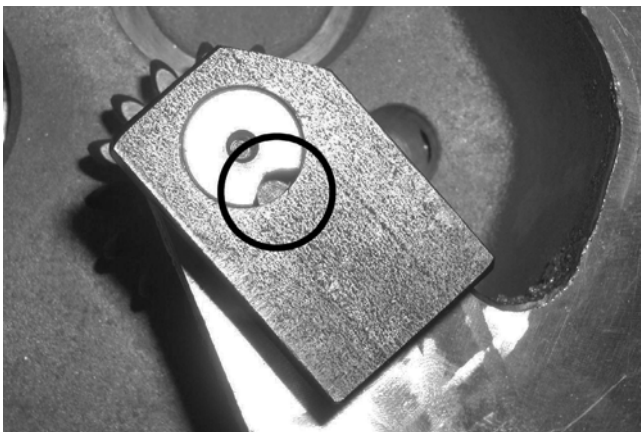




11. Install the 104-40 Dowel Pin to secure the Reverse Shift Shaft.

12. Assemble the 34-115 Reverse Shift Block to the Reverse Shift Shaft by aligning the keyway with the previously installed Dowel Pin.

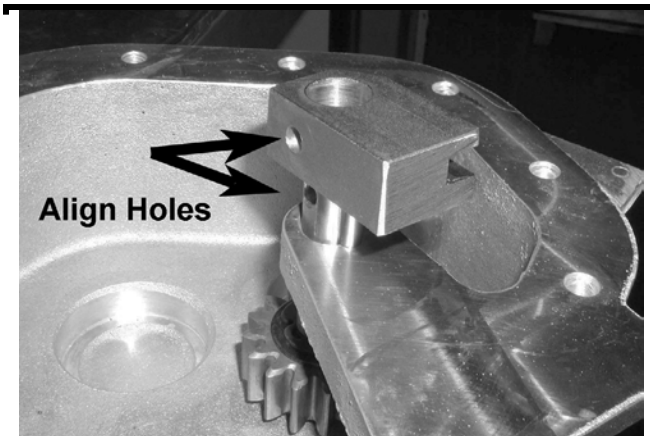


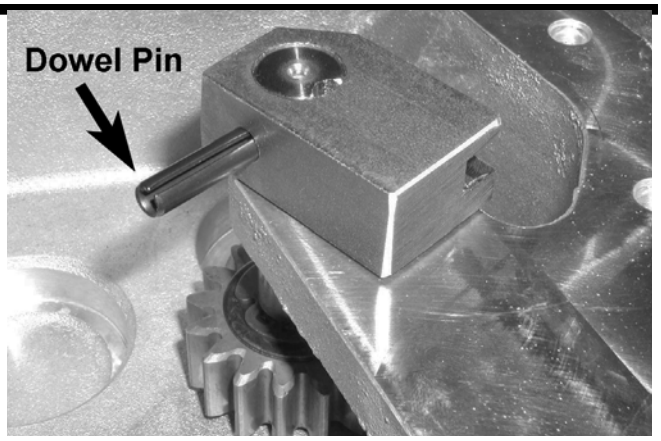


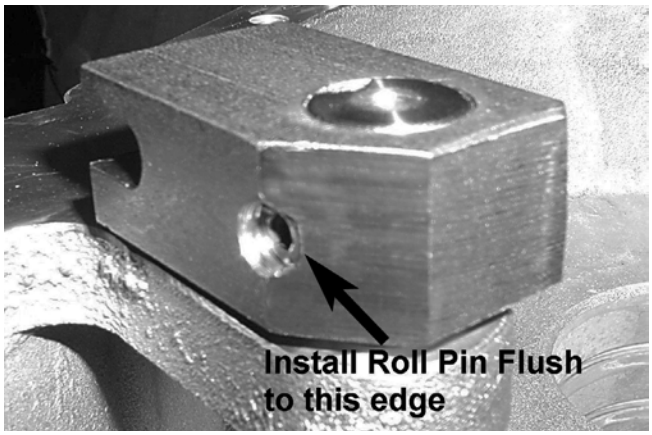
13. Tap the Shift Block on until the machined hole in both the Reverse Shift Shaft and the Shift Block are aligned, then install the 104-39 Spring Pin.

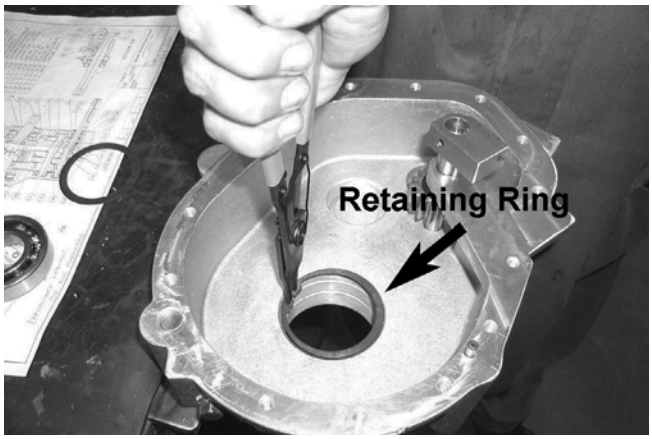
⚠ NOTE

Install the Roll Pin flush to the machine corner of the shift block as shown in Photo TR-105



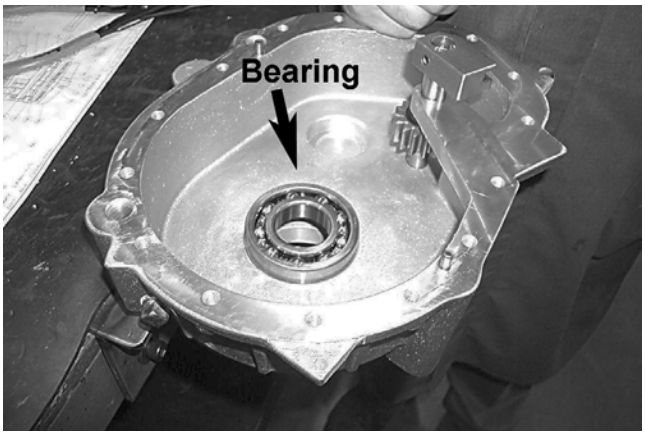






14. Install a 106-03 Retaining Ring to top groove of the Housing.

15. Press and install the 101-03 bearing into the Housing.





NOTE

If the differential consists of the high ratio 3.3 Gear Plate, it will be advantages to install the 101-01 Bearing into the cover as well at this point to avoid any interference later when installing the assembled Idler Gear Bore Assembly.

16. Press the assembled differential into the Housing.

17. Install the Idler Gear Bore Assembly at the arbor press.